

Nyle Siddiqui

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Website



LinkedIn



GitHub



Google Scholar

SUMMARY

- Currently pursuing a Ph.D. in Computer Science (Computer Vision) advised by Dr. Mubarak Shah
- NSF GRFP 2024 Honorable Mention
- B.Sc. Double Major in Computer Science and Applied Mathematics + Statistics

EDUCATION

UNIVERSITY OF CENTRAL FLORIDA

Expected Graduation: Summer 2027

PH.D STUDENT IN COMPUTER SCIENCE AT THE CENTER FOR RESEARCH IN COMPUTER VISION (CRCV)

UNIVERSITY OF WISCONSIN - EAU CLAIRE

Graduated: May 2022

B.SC IN COMPUTER SCIENCE AND APPLIED MATHEMATICS WITH STATISTICS EMPHASIS (DOUBLE MAJOR)

- GPA: 3.76/4.00
- Graduated Magna Cum Laude with both University and Departmental (Computer Science) Honors

SELECTED PAPERS

[1] DVANET: DISENTANGLING VIEW AND ACTION FEATURES FOR MULTI-VIEW ACTION RECOGNITION

Siddiqui N., Tirupattur P., Shah M. (2023). Proceedings of the AAAI Conference on Artificial Intelligence, 38(1). [Link](#)

[2] DLCR: LEVERAGING DIFFUSION AND LARGE LANGUAGE MODELS FOR CLOTHES-CHANGING RE-ID

Siddiqui N., Croitoru A., Nayak G., Shah M. (2025). IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Main Algorithms Track. **Selected for Oral Presentation.** [Link](#)

[3] VRR-QA: VISUAL RELATIONAL REASONING IN VIDEOS BEYOND EXPLICIT CUES

Swetha S., Gupta R., Siddiqui N., Kulkarni, P., Shatwell, D., Santiago, J., Fiorese, J., Shah M. (2026). IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Main Track.

[4] FLASHSSM-3D: A DISTILLED STATE SPACE MODEL FOR LIGHTNING-FAST 3D RECONSTRUCTION

Siddiqui N., Shah M. **Under review**, Neural Information Processing Systems (NeurIPS) 2026.

[5] CAMBRAIN - REAL-TIME, CONTINUOUS EEG INFERENCE WITH CAUSAL STATE SPACE MODELS

Abhliash Durgam, Siddiqui N., Jeffrey Santaigo, Qiusi Fu, Elakktat Gireesh, Shah M. **Under review**, Neural Information Processing Systems (NeurIPS) 2026.

RESEARCH EXPERIENCE

CRCV PHD STUDENT

September 2022 - Current

- Currently in my 5th year of conducting research supervised by Dr. Mubarak Shah. My main research topics include:
 - Representation Learning
 - State Space Models
 - 3D Reconstruction
 - Video Understanding
 - Generative Diffusion Models
- First-authored a paper in my first year, accepted at AAAI 2024.
- Other first-author papers in NeurIPS 2026 (2) and WACV 2025 (1).

NSF-REU PARTICIPANT - UNIVERSITY OF MINNESOTA

June 2020 - September 2020

- Conducted research in the Interactive Robotics and Vision Lab at the University of Minnesota under the supervision of Dr. Junaed Sattar
- Researched deep learning methods to calculate and mitigate risk in autonomous underwater robotic vision environments. This research experience was funded by the National Science Foundation.

TEACHING/PROFESSIONAL EXPERIENCE

AMAZON RING AI PHD INTERN

May 2026 - November 2026

- Conducted research on video object detection and grounding using state space models for Ring cameras. Paper submitted to CVPR 2027.

COMPUTER VISION PHD INTERN - METALENZ

May 2025 - September 2025

- Conducted novel research to apply vision-language models to polarimetric data for facial anti-spoofing.

NSF REU MENTOR

June 2023 - Current

- The CRCV is home to the longest running NSF-funded REU in the nation. As a PhD student, I have been a mentor to three REU students in the past 2023 and 2024 programs

UNDERGRAD RESEARCH MENTOR

September 2021 - May 2022

- Personally mentored two undergraduate researchers as they expanded on my previous research project, which lead to a publication: "Continuous User Authentication Using Machine Learning and Multi-Finger Mobile Touch Dynamics with a Novel Dataset"

AIMS WORKSHOP INSTRUCTOR

May 2021 - November 2021

- Co-instruct and co-create original material for 2-day instructional deep/machine learning-based workshops
- Workshops for participants consisting of high school students, UWEC undergraduates, and adults from the surrounding Eau Claire community. Created with the intent of contributing to the community and providing students an introduction into increasingly attractive computer science topics

UWEC MATH TUTOR

November 2018 - May 2022

Tutored undergraduate students in college-level courses, including

- College Geometry and College Algebra
- Pre-Calculus and Calculus I-III
- Statistics, Trigonometry, Linear Algebra and Probability Theory

GRANTS/AWARDS

CO-AUTHORED GRANTS

- Computer Science Student/Faculty Research Collaboration Grant (\$2,800)
- UWEC Summer Research Grant (\$3,000)
- Computer Science Student/Faculty Research Collaboration Grant (\$2,000)
- Computer Science Student/Faculty Research Collaboration Grant (\$2,000)

Fall 2021
Summer 2021
Spring 2020
Fall 2020

RECIPIENT

- UCF ORCGS Doctoral Fellowship (\$25,000/yr)
Fall 2022-27
 - Awarded to the top incoming doctoral students at the University of Central Florida. Provides guaranteed GRA funding throughout the entire PhD. Awarded by Dr. Mubarak Shah
- National Science Foundation Graduate Research Fellowship Program (NSF-GRFP) Honorable Mention
Fall 2024
 - Per NSF: "The NSF accords Honorable Mention to meritorious applicants who do not receive Fellowship offers. This is considered a significant national academic achievement."
- National Science Foundation Research Experience for Undergraduates Grant (\$9,500)
Summer 2021
 - Awarded by the NSF to conduct research at the University of Minnesota
- UWEC Diversity Mentoring Grant (\$2,000)
Fall 2021
 - Awarded for excellent undergraduate research conducted by a student from a marginalized community.